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Resolving Microporosity at Core Plug Scale: Challenges & Solutions for Permeability Estimation of Carbonate Reservoir Rocks

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Abstract

Replacing physical laboratory conventional and special core analysis experiments (CCA/SCAL) with their digital alternatives has, so far, failed to become the main stream. This is especially the case for carbonate reservoir rocks that are known for their complexity in pore types, pore size distribution and architecture. The lack of inertia towards these potentially revolutionizing concepts is due to the difficulty in reaching a micron-scale resolution. Therefore, imaging micro-porosity seems a valid justification in a delayed adoption of digital SCAL for carbonate reservoir rocks. As the industry tackles ever more challenging and lower permeability reservoirs, a wider adoption of digital SCAL becomes a more desirable objective, with significant time and cost savings compared to conventional physical experiments.

This paper leverages an extensive review of the technical literature as recent research contributed to bridging the gaps between pore space modeling and physical properties such as grain size distribution, pore size distribution, connate water saturation and plug permeability. The workflow is subdivided into two steps: (1) generating high-resolution rock images at the plug scale, and (2) estimating permeability values from these images. Resolving microporosity at the core plug scale was attempted using (1) multi-fractal behavior of the grain and pore size distribution, and (2) machine learning and pixel-based multi-point geo-statistics. For permeability estimation, multiple methods were tested including (1) graph neural network and (2) ball and stick modelling and simulation. Physical plug permeability measurements, mercury injection capillary pressure results and nuclear magnetic resonance (NMR) data (calibrated to well log NMR to account for in-situ conditions) were used for validation.

This paper shows that it is possible to embed micropores into a thresholded 3D micro-CT low-resolution volume/image at core plug scale using multi-point statistics and a high resolution 2D image from confocal microscopy as a training image. Multiple realizations were generated to quantify the residual uncertainty. However, it is not computationally feasible to run traditional pore-scale flow simulation on these high-resolution (~200 nm resolution) plug scale (inch-scale diameter) images. As an alternative for flow simulation, a graph neural network was trained to estimate permeability. The network was trained on twenty-two rock samples and produced a coefficient of determination of 0.78 and 0.87 for carbonates and

sandstones, respectively. The assessment was performed on a test dataset of rock subsamples obtained from the same formations used for training. This demonstrates the viability of using such a methodology to estimate permeability for an ensemble of realizations in an efficient manner. Overall, with advances in imaging technology, computational resources, and algorithms (including machine learning), it is feasible to perform digital SCAL at a much larger scale.

Introduction

Physical conventional and special core laboratory analyses are essential in understanding the static characteristic and dynamic behavior of rocks at the small (pore) and large (reservoir) scale. Performing these analyses can be challenging. For example, physical experiments to estimate permeability in microporous rocks are time consuming; it can take from days to weeks for one sample. With advancements in imaging technology, simulation algorithms, and computational power, digital rock analyses emerged as a viable complement, and sometimes an alternative, to physical experiments. However, practical workflows for complex rocks, such as carbonate rocks with multiple pore systems, are still not well established. This is because it is difficult to capture a representative volume of the rock with high enough magnification to resolve the needed pore structure. Furthermore, simulating the captured pore structure is difficult and intractable from a computational perspective.

Imaging/resolving micropores requires rock images at a high resolution (less than 1 micron per voxel). This resolution can be achieved only for relatively smaller samples. For digital rock simulations, a small sample might be sufficient as it is representative of the overall statistics of the rock. This is the case for some homogenous rock such as "isotropic" mudrocks (not to be confused with the layered shales). Unfortunately, this is not the case for more heterogonous rocks. In these cases, a larger sample is needed to capture the heterogeneity. Assuming a high-resolution image that resolves at the core plug scale is achievable, another problem arises. What is an efficient method to estimate permeability from such large images? The storage requirements for the raw image are too large and typical methods used such as Lattice Boltzmann simulations are not computationally efficient to run at this large scale. And so, new methods or workflows need to be developed to overcome these issues.

This paper shows that it is feasible to tackle these problems utilizing a number of tools including multi-resolution imaging, simulation, and machine learning. The paper will discuss the workflow used, shows initial results, and discusses the limitations and possible extensions and improvement to the workflow.

Methods

The general approach taken consists of two steps: 1) Acquire images at multiple scales using multiple technologies and "combine" them to arrive at a core plug volume image with resolution enough to resolve micro-porosity, and 2) estimate effective permeability for such image indirectly (as direct simulation is infeasible). Multiple directions can be taken for each step as outlined below subsections. In the paper, we will focus on specific directions as noted below.

The analysis is automated using the Python-based open-source tools for image parsing, image processing, machine learning, and visualization (van der Walt et al., 2014; Rocklin, 2015; Harris et al., 2020; Sofroniew et al., 2024). The workflow is largely automated and runs on a fairly recent high-end desktop machine.

Image Acquisition

In this study, images are obtained at multiple scales. Low resolution micro-CT scans are used to image the core plug scale, and high-resolution (micropore resolution) confocal image at the thin section scale. The images are combined to resolve the micropores at the plug scale after acquisition.

Low-resolution three-dimensional image at the core plug scale is obtained using computed tomography. Recent advances in micro-computed tomographic imaging (micro-CT) enabled imaging up to about 500 nanometers. Unfortunately, this is only feasible for smaller sample size that is typically not representative of rocks (especially rocks with larger micritized grains, micritized packestones and grainstones). Micro-CT images are obtained at different resolutions. A core plug scale image is obtained at about 15 micron per voxel resolution. Smaller millimeter to centimeter volume is obtained at about 1 to 2 micron per voxel resolution for validation purposes (although for some rocks, the micropores are smaller than this limit).

High-resolution thin section two-dimensional image is obtained using the scanning confocal microscopy as it reaches enough resolution to resolve micropores while being able to cover large representative area (Al Ibrahim et al., 2012; Shah et al., 2014; Hurley et al., 2021). Confocal microscopy uses a laser source to excite different components in the sample. The focus of the excitation in this study is the blue dye found in the epoxy of thin sections. Because the microscope enhances optical resolution and contrast by using a pinhole to eliminate out-of-focus light, resolution up to about 200 nanometer per pixel can be obtained. This is enough to image most of the micropores in typical microporous rocks. While the microscope scans the image pixel by pixel, the acquisition time per pixel is typically in microseconds, and so centimeter scale areas can be imaged at reasonable time (10s of minutes depending on the per pixel excitation time used). Acquiring a relatively large area increases the confidence in capturing the heterogeneity observed in rocks.

Quality Control

Visual quality control is performed for both the micro-CT and confocal images. In particular, confocal images obtained from thin section images are sensitive to uneven polishing producing images with uneven shading. The results can be corrected to some extent by image processing post-acquisition, but it is best to correct it by improving the sample, or modifying the acquisition workflow (e.g., by creating focus points to ensure even shading throughout the scanned area). And because the image will be used to gather spatial statistics for the spatial distribution of micropores, it is important to ensure that the imaged area is representative of the rock imaged. Representative element volume analysis (or area in the case of two-dimensional images) attempts to estimate the smallest volume (or area) that represents the rock for certain property. While not an essential analysis to the workflow, it is a good step to perform to gain more confidence in the results (Costanza-Robinson et al., 2011). Figure 1 shows the general concept behind the analysis.

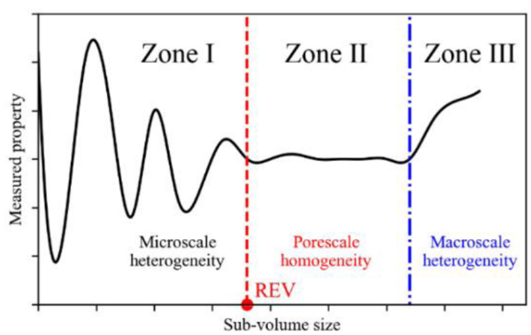

Figure 1—Representative element volume estimation conceptional model. The x-axis represents an increasing sub-sample size and the y-axis represent the measured property (Bear, 1972).

Resolving Micropores

Both the high-resolution confocal image and the low-resolution micro-CT image need to be segmented/ thresholded into different components. To do this, total porosity is needed. In this study, this is determined experimentally. Assuming the representative area analysis confirmed to some degree that the confocal image is representative of the rock, a threshold is automatically determined so that the resultant porosity value matches the experimental one. A simple search (e.g., using the bisection method) is suitable for this. Note that acquiring the image at 16bit instead of 8bit is advisable to obtain a more accurate segmentation and match the total porosity value needed. Segmenting the micro-CT image is more difficult, especially if there are no visible cutoffs observed in the intensity histogram. To estimate the porosity value that should be observed in the micro-CT image, the confocal segmented image is rescaled to the resolution of the micro-CT and the porosity value is measured. The value is used to segment the micro-CT image (by searching through for the best threshold to match the value). Another threshold is needed for the micro-CT image to segment the impermeable grains from microporous texture. Currently, this is done visually by examining the image and the intensity histogram. Figure 2 shows an example of a micro-CT image segmentation based on two defined thresholds.

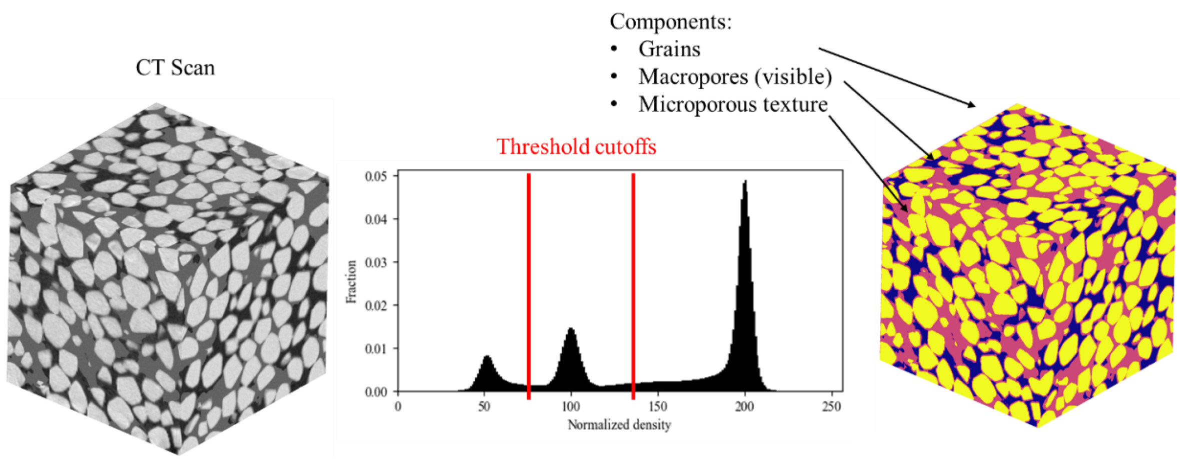

Figure 2—Thresholding of an input intensity image using defined cutoff. In this study, the micro-CT image is segmented into three components: 1) impermeable grains, 2) macropores, and 3) microporous texture.

Multipoint statistics (MPS) is used to artificially "insert" micropores into the core plug micro-CT image (Figure 3). Unlike traditional two-point geostatistics, which only considers the relationship between pairs of points, MPS takes into account the relationships between multiple points in multiple directions, allowing for a more comprehensive understanding of spatial dependencies and patterns. By incorporating training images or conceptual models, MPS can capture non-linear and non-Gaussian relationships, making it particularly useful for modeling complex systems such as porous media, geological formations, and environmental systems. In this case, areas with microporous texture in the confocal image is used as a training image. Note that depending on the thickness of the sample used in confocal microscopy, a two- or three-dimensional image can be obtained. If the obtained confocal microscopy image is two-dimensional, the spatial distribution of micropores is assumed to be isotropic. That is, we can use the same training image for all directions. The micro-CT image is resized initially to the resolution of the confocal image before simulating the micropores. Because it is sometimes infeasible to store the very large resultant image, smaller chunks from the micro-CT image are enhanced with micropores and permeability is estimated before discarded. This procedure can be done multiple times for the same chunk because MPS can produce multiple realization of the high-resolution image and the results can be used for uncertainty quantification.

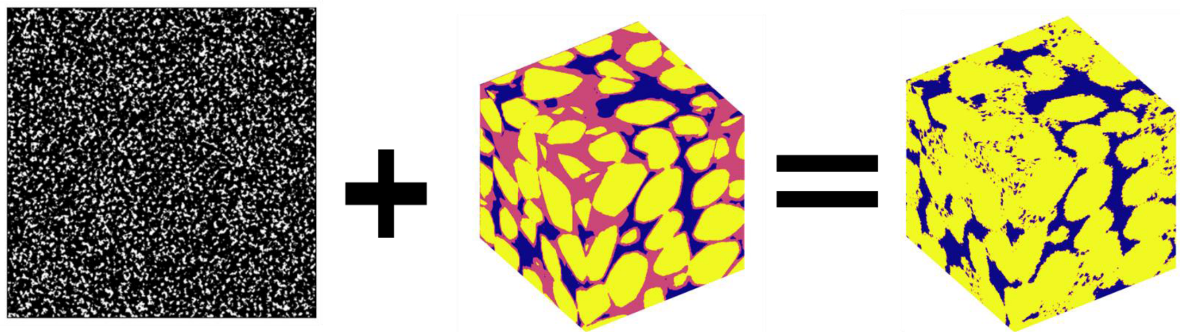

Figure 3—Workflow used to artificially insert the micropores into a lower resolution micro-CT scan image. The microporous texture (pink) is replaced with pores and grains voxels based on the simulation from multiple point statistics.

Estimating permeability

To accelerate permeability estimation, machine learning is used. Graph neural network is used to predict permeability given a pore network and grain network graphs. Figure 4 shows an example of a network architecture that was tested in this study. To prepare the input, a binarized/segmented/thresholded digital rock image is used to generate the pore and grain network graphs. Properties are estimated for both the nodes and the edges of the graphs. For example, for the grain network graph, the nodes are represented by the identified grains and the edges by the physical connection between the grains. Node (grain) properties include their size, roundness, and sphericity, and edge properties include the length of contact between the grains and the distance between their centroids. Similarly for the pore network, the pore bodies are assigned as nodes and pore throats are assigned as edges for the graph. The separation is done using the watershed algorithm where the contact between objects (pore bodies) separated are assigned as pore throats. Node (pore body) properties include their size, and sphericity and edge (pore throat) properties include the length of contact between the pore bodies.

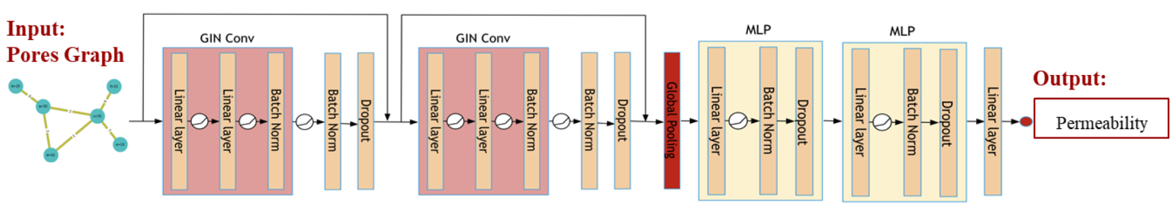

Figure 4—An example of graph neural network architecture tested in this study.

Results

Figure 5 shows an example of a slice of a micro-CT scan (left) and a realization where microporosity was inserted using multiple point statistics. Note that: 1) the process is applied on the three-dimensional volume of scaled down micro-CT image, and 2) multiple realizations can be generated.

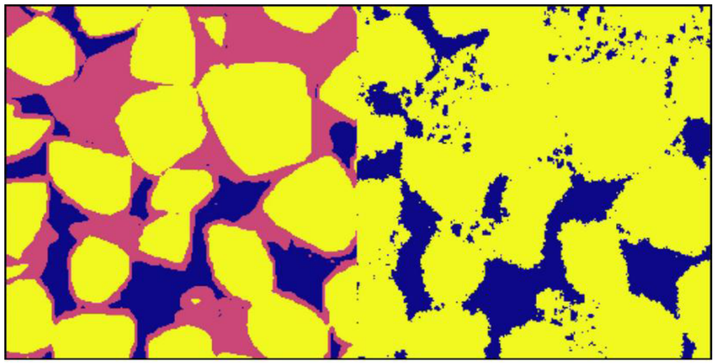

Figure 5—Example of micro-pores insertion (right) on synthetic image (left). The microporous texture (pink) is replaced with pores and grains voxels based on the simulation from multiple point statistics.

To estimate permeability in an efficient manner, multiple graph neural networks were trained on a micro-CT dataset. The dataset (Figure 6) was generated by Liu et al. (2023). Figure 7 and Figure 8 show the results for sandstones and carbonate rocks respectively. Overall, results show that it is possible to train models to predict permeability if the training dataset contains similar rocks. However, it is more challenging to predict the permeability of new rock types or formations, especially for carbonates. This is expected as machine learning models are mainly suitable for interpolation only, not extrapolation.

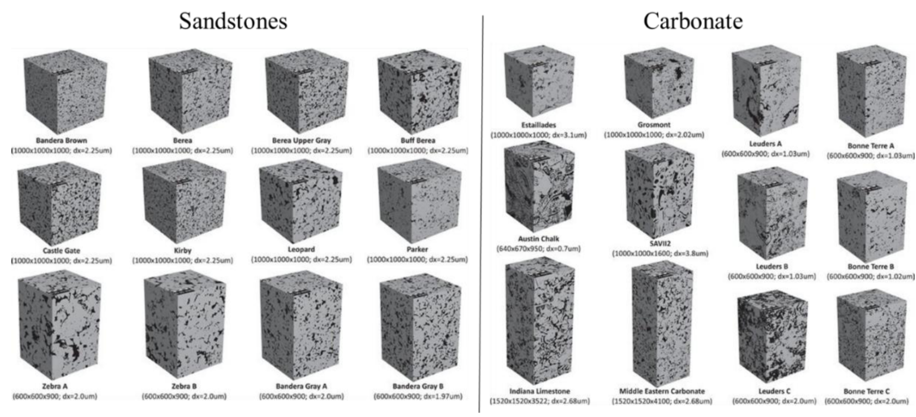

Figure 6—Dataset used for training the graph neural network model to predict the absolute permeability. The permeability estimated for the volumes is calculated using Lattice Boltzmann. Figure is modified from (Liu et al., 2023).

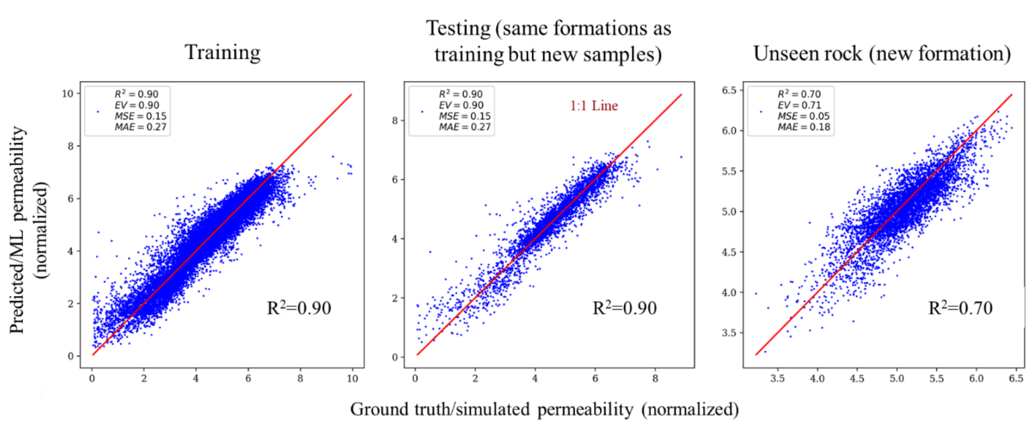

Figure 7—Graph neural network results for sandstone dataset.

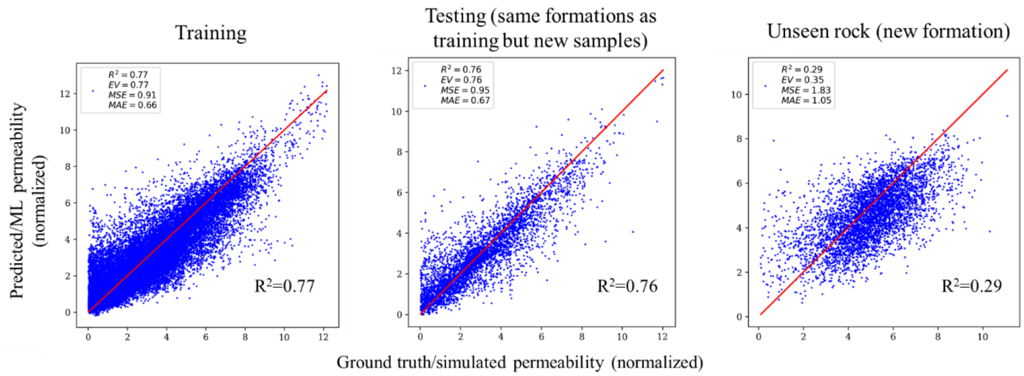

Figure 8—Graph neural network results for carbonate dataset.

Discussion

Results for synthetically inserting micropores into micro-CT imaging are promising but it can be improved. For example, the issue of the best thresholding methods for both micro-CT and confocal microscopy images can further be studied. The different rock components in micro-CT images (impermeable grains, microporous texture, and macro pores) overlap in their intensity as observed in the histogram. A simplification is made to select a value to threshold between two components. Other advanced methods can be used, for example, other studies showed that fitting Gaussian distribution on the components produces better segmentation. The value chosen for the segmentation of confocal image is another. The value is chosen by matching the experimental porosity obtained from the core plug. This assumes that the core plug and the confocal image are both representative of the rock. Ensuring that the confocal microscopy image is representative of the core plug is another issue. Currently, the confocal image is analyzed separately to determine homogeneity. However, if a confocal image is determined to be homogenous, this does not guarantee that it is representative of the full core plug. This is because there might be some heterogeneity at a larger (plug scale) that is not captured (see Figure 1, zone III). And so, an analysis that considers both images is needed when deciding if the confocal image is representative of the core plug.

Representing pore networks as graph is desirable because: 1) storage requirement is relatively small compared to the original voxel image, and 2) there is potential to connect graphs together to obtain a representation at the core plug scale. However, utilizing machine learning models for permeability estimation is risky. Guardrails must be set up to ensure that the results can be trusted. For example, similarity metrics can be used to determine if a new image is sufficiently similar to the images in the training data. This is because, generally, machine learning models are good for interpolation, but they are not great at extrapolation. Furthermore, comparison between simulation and model output for a few samples for new datasets should be made occasionally. Finally, new formations and rock types that do not exist in the training dataset of the machine learning model should be incorporated and the model should be finetuned continuously.

Conclusions and Final Remarks

The results of this study have shown that a combination of imaging hardware and computer vision algorithms (including machine learning) improvements can enable more accurate and practical property estimation from digital rocks. Multifaceted approach must be taken to tackle the issue. For example, integration with other types of data, if available, is desirable. For example, nuclear magnetic resonance (NMR) can give indication of the pore size distribution which can assist in the segmentation of the core plug. In addition, improving the resolution of micro-CT scan images using machine learning is possible (instead of using multiple point statistics). Finally, using machine learning for direct estimation of permeability from low-resolution three-dimensional micro-CT images, or high-resolution two-dimensional confocal images is an interesting direction.

Overall, the existence of training datasets (digital rock images and their corresponding physical properties, either physically measured or simulated) would help the industry build better machine learning models. In this regard, synthetic realistic digital rocks (Al Ibrahim et al., 2019) might play a role in pretraining machine learning models using large datasets before being finetuned on smaller real rock datasets.

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